# GRIDMIND

## Autonomous Smart Energy Load Balancing & Demand Response System

### A Multi-Agent System Proof-of-Concept

*Developed in Partnership with ECG & GRIDCo*

DCIT 403 — Designing Intelligent Agents
Department of Computer Science, University of Ghana, Legon

# 1. Project Overview

For over two decades, Ghana's power crisis — colloquially known as dumsor, meaning "off and on" in Akan — has drained an estimated $2.1 billion annually from the national economy, shuttered businesses, destabilised hospitals, and exhausted ordinary Ghanaians who have built entire domestic routines around unpredictable outages. The Electricity Company of Ghana (ECG) and the Ghana Grid Company (GRIDCo) manage a national peak demand of approximately 2,800–3,200 MW against an installed generation capacity that is chronically undermined by low Akosombo reservoir levels, aging transmission infrastructure, and reactive — rather than intelligent — load management.

GridMind is a multi-agent system designed to address this coordination failure directly. It deploys six classes of intelligent agents — each owning a specific slice of the grid management problem — to monitor real-time demand across five of Ghana's highest-load districts, negotiate energy procurement from domestic renewable installations, and orchestrate voluntary demand reduction with named industrial consumers, all before forced load shedding becomes necessary. This system is framed as a proof-of-concept pilot for potential adoption by ECG's Central Dispatch Centre and GRIDCo's System Control Centre.

## 1.1 The Five Pilot Districts

GridMind's pilot scope covers five districts that together represent the most economically critical and load-intensive sections of Ghana's distribution network. The table below reflects realistic peak demand estimates based on district character and available ECG load data:

| District | Character | Peak Demand (MW) | ECG Feeder Region |
|---|---|---|---|
| Tema Industrial Enclave | Heavy industry, port, refinery, cement | 350–400 MW | Eastern Corridor |
| Accra Central (Ministries / Ridge) | Government, commercial CBD, hospitals | 160–190 MW | Greater Accra North |
| Kumasi Suame Magazine | Auto/metal industry, dense commercial | 130–150 MW | Ashanti Region |
| Kasoa Corridor | High-density residential, small commerce | 80–100 MW | Central/Greater Accra |
| Takoradi Harbour District | Port ops, oil support, industrial estate | 150–175 MW | Western Region |

*Pilot total managed load: approximately 870–1,015 MW — roughly 30% of Ghana's national peak demand. This scope is intentionally bounded to what a single proof-of-concept deployment could realistically instrument and manage.*

## 1.2 System Objectives

- Detect demand anomalies across all five districts within one simulation tick (≤ 500ms) of occurrence
- Predict load surges at least 5 ticks ahead using belief-based forecasting, enabling preemptive procurement
- Autonomously negotiate energy injection from Kaleo Solar, Nzema Solar, and Keta Coastal Wind via FIPA Contract Net Protocol
- Negotiate voluntary demand reduction with VALCO, Tema Oil Refinery, GHACEM, and other named industrial consumers before activating dumsor rotation
- Produce a fully auditable decision log — every negotiation, every intention revision, every load shed — timestamped and attributed to a specific agent and reason code
- Demonstrate measurable improvement in grid stability over ECG's current threshold-only baseline

## 1.3 Course Concepts Coverage Map

| Concept | GridMind Implementation | Primary Agent(s) | Depth |
|---|---|---|---|
| BDI Architecture | ECG Central Dispatch agent has explicit typed Beliefs, priority-ordered Desires, and an intention reconsideration loop | ecg_central_dispatch | Deep / Showpiece |
| Prometheus Methodology | Full 3-phase design: system spec (goals, scenarios, roles) → architectural design (agents + interaction diagram) → detailed design (per-agent plan libraries) | All agents | Complete |
| FIPA-ACL Communication | CFP, PROPOSE, ACCEPT-PROPOSAL, REJECT-PROPOSAL, REFUSE, REQUEST, AGREE, FAILURE, INFORM, BROADCAST all in active use | All agents | Full Coverage |
| Reactive Behavior | GRIDCo Fault Watch and district zone agents are pure sense-act agents with no internal deliberation | gridco_fault_watch, zone agents | Clear Contrast |
| Deliberative Behavior | ECG Forecast Unit reasons over a 20-tick rolling history before advising; ECG Central Dispatch reasons over its full belief base before committing to an intention | ecg_forecast_unit, ecg_central_dispatch | Deep |
| Multi-Agent Coordination | Full Contract Net Protocol between ECG Central Dispatch and three renewable source agents | ecg_central_dispatch + renewables | Full Protocol |
| Dynamic Environments | Mid-simulation event injection: low-water curtailment, Spintex feeder fault, shift-change spike, GRIDCo line trip | event_injector coroutine | 5 Named Scenarios |

| | | | |
|---|---|---|---|
| System Evaluation | MAS vs. ECG baseline comparison across 5 KPIs with hard numbers | metrics_collector + baseline_runner | Quantitative |

# 2. Prometheus Methodology

The Prometheus methodology provides a structured, three-phase engineering process for agent-oriented systems. GridMind follows all three phases with explicit design artifacts at each stage. This section constitutes the primary Prometheus documentation deliverable for the project.

## 2.1 Phase 1 — System Specification

### 2.1.1 System Goals

- G1 — Stability: Maintain managed grid load within 95% of combined district capacity at all times during the simulation
- G2 — Speed: Detect and initiate a coordinated response to any demand anomaly within 3 simulation ticks of its occurrence
- G3 — Renewable Priority: Exhaust all available renewable procurement options before activating demand response or load shedding
- G4 — Minimise Dumsor: Reduce the total number of ticks during which any district enters a load-shedding state, compared to ECG's current threshold-only baseline
- G5 — Auditability: Record every agent decision, intention revision, and FIPA negotiation outcome in a timestamped, human-readable audit log compatible with ECG's reporting requirements
- G6 — Graceful Degradation: When all procurement and demand response options are exhausted, execute controlled, prioritised load shedding — lowest economic impact first — rather than uncontrolled blackout

### 2.1.2 Named Scenarios

#### 🌊 Scenario 1: Akosombo Low-Water Season Load Curtailment

VRA reduces Akosombo hydro output by up to 40% during the dry season (typically January–March), removing approximately 300–500 MW from the national supply mix. ECG Central Dispatch agent detects a sustained drop in available renewable bids from Bui Hydro, adjusts its total_available_supply belief downward, and proactively shifts its desire from OPTIMIZE_RENEWABLES to PREEMPTIVE_RESPONSE. The Forecast Unit's predictions become critical here — it must detect the supply-side erosion before peak demand hours arrive.
Agents exercised: ecg_central_dispatch (BDI revision), bui_hydro (REFUSE pattern), ecg_forecast_unit (supply-side forecasting)

#### 🏭 Scenario 2: Tema Heavy Industrial Simultaneous Shift-Change Spike

At 06:00 and 18:00 each day, Tema Industrial Enclave experiences a documented demand spike of 60–80 MW as shift changes occur simultaneously across Meridian Port Services, Tema Oil Refinery, and VALCO. The FaultDetectorAgent (GRIDCo Fault Watch) detects the spike within one tick. ECG Central Dispatch

reconsiders its current intention and initiates a Contract Net round with Kaleo Solar and Nzema Solar. This scenario demonstrates predictable reactive response.
Agents exercised: tema_industrial zone agent (spike reporting), gridco_fault_watch (anomaly detection), ecg_central_dispatch (CNP initiation)

### 🌑 Scenario 3: Dumsor Rotation Schedule Activation

When all renewable procurement and demand response options have been exhausted and total demand exceeds 98% of managed capacity, ECG Central Dispatch activates a structured dumsor rotation schedule. Rather than arbitrary shedding, the agent's load_shedding_plan prioritises by district economic weight: Kasoa Corridor (residential) is shed before Accra Central, which is shed before Tema Industrial (critical industrial and port operations are the last resort). Each rotation is logged with a reason code.
Agents exercised: ecg_central_dispatch (load_shedding_plan), accra_industrial_dr (final DR report), all zone agents (BROADCAST receipt)

### ⚠️ Scenario 4: ECG Feeder Cable Fault — Spintex Road Corridor

A fault on the Spintex Road feeder cable causes Accra Central's reported demand to drop suddenly to zero — not because demand fell, but because the sensor feed is lost. GRIDCo Fault Watch flags this as an anomaly (a drop, not a spike) and sends an INFORM with fault_type = 'feeder_loss'. ECG Central Dispatch must distinguish a feeder fault from a genuine demand reduction — a subtlety handled by checking whether neighbouring districts show compensating demand increases. If confirmed, the agent isolates the zone from its belief base and continues managing the remaining four districts.
Agents exercised: gridco_fault_watch (fault classification), ecg_central_dispatch (belief isolation logic), ecg_forecast_unit (forecast adjustment)

### 🔗 Scenario 5: GRIDCo Inter-Regional Transmission Line Trip

A trip on GRIDCo's Volta-Kumasi 161kV line cuts power transfer capacity to Kumasi Suame by approximately 90 MW. The Kumasi zone agent immediately reports a capacity_ceiling_mw revision downward. ECG Central Dispatch must revise its total_capacity belief, re-evaluate all active intentions, and cascade a REQUEST to accra_industrial_dr for emergency demand reduction from GHACEM Takoradi and other Western Region consumers. This scenario exercises cascading belief revision and multi-agent emergency coordination.
Agents exercised: kumasi_suame zone agent (capacity revision), ecg_central_dispatch (cascading belief update + intention reconsideration), accra_industrial_dr (emergency DR)

## 2.1.3 System Roles

| Role | Description | Mapped To |
|---|---|---|
| District Monitor | Continuously read and broadcast demand for one district. Detect local trend direction. | One ZoneAgent per district (×5) |

| | | |
|---|---|---|
| | Send readings to both ECG Central Dispatch and GRIDCo Fault Watch. | |
| Grid Coordinator | Maintain system-wide situational awareness. Select and execute the highest-priority response intention. Orchestrate all negotiations. | ECG Central Dispatch Agent (×1) |
| Renewable Supplier | Report available generation capacity each tick. Respond to CFP with a PROPOSE or REFUSE. Confirm injection on ACCEPT-PROPOSAL. | Kaleo Solar, Nzema Solar, Keta Wind Agents (×3) |
| Demand Manager | Negotiate voluntary load reduction with named industrial consumers. Report achieved reduction back to Grid Coordinator. | Accra Industrial DR Agent (×1) |
| Fault Watcher | Monitor all incoming demand readings for statistical anomalies. Fire INFORM alerts to Grid Coordinator on detection. | GRIDCo Fault Watch Agent (×1) |
| Forecaster | Build rolling demand history per district. Compute short-horizon load forecasts. Alert Grid Coordinator to predicted surges before they occur. | ECG Forecast Unit Agent (×1) |

## 2.2 Phase 2 — Architectural Design

### 2.2.1 Agent Roster

| Agent Name (JID) | Type | Count | SPADE Behaviour |
|---|---|---|---|
| tema_industrial@gridmind.gh | Reactive ZoneAgent | 1 | PeriodicBehaviour |
| accra_central@gridmind.gh | Reactive ZoneAgent | 1 | PeriodicBehaviour |
| kumasi_suame@gridmind.gh | Reactive ZoneAgent | 1 | PeriodicBehaviour |
| kasoa_corridor@gridmind.gh | Reactive ZoneAgent | 1 | PeriodicBehaviour |
| takoradi_harbour@gridmind.gh | Reactive ZoneAgent | 1 | PeriodicBehaviour |
| ecg_central_dispatch@gridmind.gh | BDI Deliberative | 1 | CyclicBehaviour + MessageReceiveBehaviour |
| kaleo_solar@gridmind.gh | Deliberative Reactive | 1 | MessageReceiveBehaviour |
| nzema_solar@gridmind.gh | Deliberative Reactive | 1 | MessageReceiveBehaviour |
| keta_wind@gridmind.gh | Deliberative Reactive | 1 | MessageReceiveBehaviour |
| accra_industrial_dr@gridmind.gh | Deliberative | 1 | MessageReceiveBehaviour |
| gridco_fault_watch@gridmind.gh | Purely Reactive | 1 | MessageReceiveBehaviour |

| | | | |
|---|---|---|---|
| ecg_forecast_unit@gridmind.gh | Deliberative | 1 | PeriodicBehaviour |

### 2.2.2 Agent Interaction Specification

| From Agent | To Agent | Performative | Content Summary |
|---|---|---|---|
| tema_industrial (+ 4 others) | ecg_central_dispatch | INFORM | { district, demand_mw, capacity_mw, trend, tick } |
| tema_industrial (+ 4 others) | gridco_fault_watch | INFORM | { district, demand_mw, tick } |
| gridco_fault_watch | ecg_central_dispatch | INFORM | { district, fault_type, severity_sigma, tick } |
| ecg_forecast_unit | ecg_central_dispatch | INFORM | { district, forecast_mw, horizon_ticks, confidence } |
| ecg_central_dispatch | kaleo_solar / nzema_solar / keta_wind | CFP | { request_mw, deadline_ticks, conversation_id } |
| kaleo_solar / nzema_solar / keta_wind | ecg_central_dispatch | PROPOSE | { source, available_mw, cost_coeff, conversation_id } |
| kaleo_solar / nzema_solar / keta_wind | ecg_central_dispatch | REFUSE | { source, reason, conversation_id } |
| ecg_central_dispatch | winning renewable | ACCEPT-PROPOSAL | { conversation_id, accepted_mw } |
| ecg_central_dispatch | losing renewables | REJECT-PROPOSAL | { conversation_id } |
| ecg_central_dispatch | accra_industrial_dr | REQUEST | { target_reduction_mw, priority, tick } |
| accra_industrial_dr | ecg_central_dispatch | AGREE | { acknowledged: true } |
| accra_industrial_dr | ecg_central_dispatch | INFORM | { achieved_mw, consumers_engaged, tick } |
| accra_industrial_dr | ecg_central_dispatch | FAILURE | { achieved_mw, gap_mw, reason } |
| ecg_central_dispatch | all zone agents | BROADCAST | { system_status, active_intention, tick, audit_ref } |

### 2.2.3 Environment Model

The shared environment is a Python dataclass instantiated once in main.py and passed by reference to all agents at startup. No agent writes to another agent's slice; each agent owns and updates only its designated fields:

```
@dataclass
class GhanaGridEnvironment:
    # --- District State (written by ZoneAgents) ---
    district_demand_mw: Dict[str, float]       # keyed by district JID prefix
    district_capacity_mw: Dict[str, float]     # static ceiling per district
    district_status: Dict[str, str]            #
'normal'|'warning'|'shedding'|'fault'

    # --- Renewable State (written by RenewableAgents) ---
    renewable_available_mw: Dict[str, float]   # keyed by source JID prefix
    renewable_injecting: Dict[str, bool]

    # --- Simulation Control ---
    tick: int = 0
    event_queue: List[dict] = field(default_factory=list)
    audit_log: List[dict]   = field(default_factory=list)
```

# 3. Agent Specifications

This section constitutes Phase 3 of the Prometheus methodology — detailed design. Each agent is specified with its internal belief structure, plan library, message interface, and behavior classification. All identifiers and values use real Ghana grid parameters.

## 3.1 District Zone Agents (×5)

**Agent Profile — All Five District Agents**

```
Type:        Purely Reactive
JIDs:        tema_industrial@gridmind.gh
             accra_central@gridmind.gh
             kumasi_suame@gridmind.gh
             kasoa_corridor@gridmind.gh
             takoradi_harbour@gridmind.gh
Behaviour:   PeriodicBehaviour — fires every TICK_INTERVAL seconds
Role:        District Monitor
```

### Per-District Configuration

| District Agent | Base Demand (MW) | Peak Ceiling (MW) | Noise Factor | Typical Trend Pattern |
|---|---|---|---|---|
| tema_industrial | 290 MW | 400 MW | ±3% | Twin spikes at 06:00 and 18:00 shift changes |
| accra_central | 140 MW | 190 MW | ±2% | Business-hours ramp, sharp evening dip |
| kumasi_suame | 110 MW | 150 MW | ±2.5% | Gradual morning ramp, flat midday, soft evening peak |
| kasoa_corridor | 65 MW | 100 MW | ±4% | Residential evening peak (19:00–22:00), low daytime |
| takoradi_harbour | 125 MW | 175 MW | ±2% | Port operations flat load, oil-industry night shift adds 20MW |

### Belief Structure

```
# Shared across all 5 district agents — values differ per district
beliefs = {
    'district_id':      'tema_industrial',   # unique per agent
    'current_mw':        290.0,              # updated each tick
    'capacity_mw':       400.0,              # static ceiling
```

```
    'utilisation_pct':   0.725,              # current_mw / capacity_mw
    'last_5_readings':   [],                 # rolling window for trend
    'trend':             'stable',           # 'rising'|'falling'|'stable'
    'status':            'normal',           # set by BROADCAST from ECG Dispatch
}
```

### Plan Library

| Plan | Trigger | Body |
|---|---|---|
| read_and_broadcast | Periodic tick | Sample demand from env.district_demand_mw, apply Gaussian noise at district noise_factor, send INFORM to ecg_central_dispatch and gridco_fault_watch, update beliefs |
| compute_local_trend | After each read | Append reading to last_5_readings (cap at 5); compute linear slope; set trend belief to 'rising', 'falling', or 'stable' |
| receive_system_status | BROADCAST arrives from ecg_central_dispatch | Update local status belief with broadcast content; log to console dashboard |

## 3.2 ECG Central Dispatch Agent — BDI Showpiece

**Agent Profile — ecg_central_dispatch@gridmind.gh**

```
Type:        BDI Deliberative Agent — the system's cognitive centrepiece
JID:         ecg_central_dispatch@gridmind.gh
Institution: Models the ECG Central Dispatch Centre, Accra
Behaviours:  BDILoopBehaviour (CyclicBehaviour) + FIPAReceiveBehaviour
             + BroadcastBehaviour (PeriodicBehaviour every 5 ticks)
```

### Belief Base

```
@dataclass
class ECGCentralDispatchBeliefs:
    # --- Demand Beliefs ---
    district_demands: Dict[str, float] = field(default_factory=lambda: {
        'tema_industrial': 0.0, 'accra_central': 0.0,
        'kumasi_suame': 0.0, 'kasoa_corridor': 0.0, 'takoradi_harbour': 0.0
    })
    total_managed_demand_mw: float = 0.0
    total_managed_capacity_mw: float = 1015.0  # sum of 5 district ceilings

    # --- Supply Beliefs ---
    renewable_bids: Dict[str, float] = field(default_factory=lambda: {
        'kaleo_solar': 0.0, 'nzema_solar': 0.0, 'keta_wind': 0.0
    })
    total_renewable_available_mw: float = 0.0
```

```
    # --- Situation Beliefs ---
    active_faults: List[str] = field(default_factory=list)
    forecast: Dict[str, float] = field(default_factory=dict)
    demand_response_active: bool = False
    pending_cnp_id: Optional[str] = None
    pending_cnp_proposals: List[dict] = field(default_factory=list)

    # --- Audit ---
    audit_log: List[dict] = field(default_factory=list)
```

## Desire Hierarchy (Priority-Ordered)

Desires are evaluated top-to-bottom. The first condition that evaluates to True determines the active desire. This list encodes ECG Central Dispatch's operational priorities:

| Priority | Desire Name | Activation Condition | Real-World Meaning |
|---|---|---|---|
| 1 — Highest | EMERGENCY_STABILISE | total_managed_demand_mw > total_managed_capacity_mw × 0.98 | Grid is at 98% capacity — imminent blackout risk |
| 2 | RESPOND_TO_FAULT | len(active_faults) > 0 | GRIDCo Fault Watch has flagged at least one district anomaly |
| 3 | PREEMPTIVE_RESPONSE | any district in forecast dict exceeds 85% of its capacity ceiling within 5 ticks | ECG Forecast Unit predicts a surge — act before it materialises |
| 4 | OPTIMISE_RENEWABLES | total_renewable_available_mw > 0 AND demand is trending upward | Renewable headroom exists — maximise clean energy use |
| 5 — Lowest | MONITOR | All conditions above are False | Normal operations — observe and maintain situational awareness |

## Intention Selection & Plan Mapping

| Intention | Triggered By Desire | Plan Executed | Reconsideration Trigger |
|---|---|---|---|
| initiate_contract_net | EMERGENCY_STABILISE or PREEMPTIVE_RESPONSE | contract_net_plan | FAULT INFORM mid-CNP → abandon, adopt RESPOND_TO_FAULT |
| emergency_contract_net | RESPOND_TO_FAULT | contract_net_plan (compressed deadline) | Second fault INFORM → escalate to EMERGENCY_STABILISE |

| | | | |
|---|---|---|---|
| trigger_demand_response | EMERGENCY_STABILISE after CNP fails or returns insufficient MW | demand_response_plan | FAILURE from accra_industrial_dr → activate load_shedding_plan |
| activate_dumsor_rotation | EMERGENCY_STABILISE after both CNP and DR fail | load_shedding_plan | Demand drops below 90% → de-escalate to MONITOR |
| optimise_renewable_mix | OPTIMISE_RENEWABLES | renewable_optimisation_plan | Fault INFORM → reconsider immediately |
| passive_monitoring | MONITOR | monitoring_plan | Any higher desire activates → suspend and switch |

## Plan Library — Full Detail

| Plan | Full Description |
|---|---|
| monitoring_plan | Receive INFORM messages from all 5 district agents and update district_demands beliefs. Recompute total_managed_demand_mw. Check all desire thresholds. If all thresholds clear, log a 'system_normal' entry to audit_log. Send periodic BROADCAST digest to all zone agents every 5 ticks. |
| contract_net_plan | Generate a UUID conversation_id. Send CFP to kaleo_solar, nzema_solar, and keta_wind with request_mw = (total_managed_demand_mw - total_managed_capacity_mw × 0.90) and deadline_ticks = CNP_DEADLINE_TICKS. Collect PROPOSE messages into pending_cnp_proposals buffer, keyed by conversation_id. After deadline: score each proposal as (available_mw / cost_coeff). Send ACCEPT-PROPOSAL to highest scorer, REJECT-PROPOSAL to all others. On REFUSE from all sources: fall through to demand_response_plan. Log full negotiation trace to audit_log. |
| demand_response_plan | Send REQUEST to accra_industrial_dr specifying target_reduction_mw. Await AGREE acknowledgement. Await INFORM result (actual MW shed) or FAILURE (gap MW). Update demand_response_active belief. If FAILURE and gap_mw > 50: activate load_shedding_plan for the gap. Log to audit_log with ECG reporting format. |
| load_shedding_plan | Determine shedding order by district economic weight (lowest impact first): kasoa_corridor → accra_central → kumasi_suame → takoradi_harbour → tema_industrial (last resort, never fully shed). Update env.district_status to 'shedding'. Send BROADCAST to all agents with reason_code = 'DUMSOR_ROTATION'. Log with timestamp, districts shed, MW recovered, reason chain. |
| renewable_optimisation_plan | Initiate a soft CFP (no emergency threshold) to all three renewable agents. Accept the cheapest available bid. Update renewable_bids belief. Log MW procured and cost coefficient delta vs. previous tick. This plan runs when capacity headroom exists and demand is rising — proactive rather than reactive. |
| intention_reconsideration | Triggered by any FAULT INFORM or sudden desire threshold crossing mid-plan. Calls abandon_current_intention(), which logs the interrupted plan with tick and reason. |

| | |
|---|---|
| | Re-evaluates the desire stack. Adopts the new highest-priority intention. This is the BDI reconsideration mechanism — make it visible in the demo. |

## 3.3 Renewable Source Agents (×3)

### 3.3.1 Kaleo Solar Plant — kaleo_solar@gridmind.gh

**Real-World Reference: Kaleo Solar Plant**

```
Location:           Kaleo, Wa, Upper West Region of Ghana
Installed Capacity: ~155 MW (one of West Africa's largest utility-scale solar
farms)
Operator:           BXC Company Ltd / Government of Ghana
Realistic Output:   Peak generation 140-155 MW (10:00-15:00); drops to ~0 after
18:30
GridMind Model:     available_mw = 155 * weather_factor * time_of_day_factor
```

```
beliefs = {
    'source_id':          'kaleo_solar',
    'installed_mw':        155.0,
    'available_mw':        0.0,          # computed each tick
    'cost_coeff':          0.28,          # GHS/MWh proxy (low marginal cost)
    'weather_factor':      1.0,           # 0.05-1.0 (cloud cover events)
    'time_of_day_factor':  0.0,           # 0.0 (night) to 1.0 (solar noon)
    'currently_injecting': False,
}
```

### 3.3.2 Nzema Solar Plant — nzema_solar@gridmind.gh

**Real-World Reference: Nzema Solar Plant**

```
Location:           Atuabo, Ellembelle District, Western Region of Ghana
Installed Capacity: ~20 MW (Ghana's earliest large utility solar project,
operational since 2015)
Operator:           Blue Energy (now BXC)
GridMind Model:     available_mw = 20 * weather_factor * time_of_day_factor
Note:               Smaller plant — typically used to top up after Kaleo
allocation
```

### 3.3.3 Keta Coastal Wind — keta_wind@gridmind.gh

**Real-World Reference: Accra–Keta Coastal Wind Corridor**

```
Location:           Keta Peninsula and Ada Foah coastal strip, Greater Accra /
Volta Region
```

```
Modelled Capacity: ~50 MW (representative of planned coastal wind development
in Ghana)
Realistic Output:  Strongest at night and early morning; drops midday
GridMind Model:    available_mw = 50 * wind_speed_factor (inverse of solar peak
hours)
Note:              Complements solar — wind peaks when solar dips, useful for
night-time grid support
```

### Shared Plan Library (All Renewable Agents)

| Plan | Trigger | Body |
|---|---|---|
| update_capacity | PeriodicBehaviour each tick | Compute available_mw using weather_factor and time_of_day_factor. Update env.renewable_available_mw[source_id]. |
| respond_to_cfp | CFP received from ecg_central_dispatch | If available_mw > 5.0: send PROPOSE with available_mw and cost_coeff and conversation_id. Else: send REFUSE with reason='insufficient_capacity'. |
| confirm_injection | ACCEPT-PROPOSAL received | Set currently_injecting = True. Update available_mw -= accepted_mw. Send INFORM confirm. Schedule injection_end event after N ticks. |
| end_injection | injection_end event fires | Set currently_injecting = False. Restore available_mw. Send INFORM to ecg_central_dispatch with reason='injection_complete'. |

## 3.4 Accra Industrial Demand Response Agent

### Agent Profile — accra_industrial_dr@gridmind.gh

```
Type:        Deliberative Agent
JID:         accra_industrial_dr@gridmind.gh
Institution: Models ECG's Industrial Demand Response Programme
Behaviour:   MessageReceiveBehaviour
```

### Managed Industrial Consumer Roster

| Consumer | Location | Contracted DR Capacity (MW) | Compliance Rate | Priority |
|---|---|---|---|---|
| VALCO (Volta Aluminium Company) | Tema | 80 MW | 0.95 | 1st — Largest capacity |
| Tema Oil Refinery | Tema Industrial Enclave | 35 MW | 0.88 | 2nd |

| | | | | |
|---|---|---|---|---|
| Meridian Port Services Tema | Tema Port | 20 MW | 0.80 | 3rd |
| Ghana Cement (GHACEM) Takoradi | Takoradi | 25 MW | 0.82 | 4th |
| Accra Brewery Limited | Achimota Industrial | 15 MW | 0.75 | 5th — Smallest |

### Belief Structure

```
beliefs = {
    'consumer_roster': {
        'VALCO':              {'contracted_dr_mw': 80, 'compliance_rate': 0.95,
'active': False},
        'tema_oil_refinery':  {'contracted_dr_mw': 35, 'compliance_rate': 0.88,
'active': False},
        'meridian_port':      {'contracted_dr_mw': 20, 'compliance_rate': 0.80,
'active': False},
        'GHACEM_takoradi':    {'contracted_dr_mw': 25, 'compliance_rate': 0.82,
'active': False},
        'accra_brewery':      {'contracted_dr_mw': 15, 'compliance_rate': 0.75,
'active': False},
    },
    'total_contracted_mw': 175,
    'current_reduction_mw': 0.0,
    'pending_request': None,
}
```

### Negotiation Logic

The DR agent iterates consumers in descending compliance_rate order, simulating an acceptance draw against each consumer's compliance_rate probability. It accumulates achieved MW until the target is met or the roster is exhausted:

```
def negotiate_reduction(self, target_mw: float) -> float:
    achieved_mw = 0.0
    for consumer_id, data in sorted_by_compliance(self.beliefs['consumer_roster']):
        if achieved_mw >= target_mw:
            break
        accepted = random() < data['compliance_rate']
        if accepted:
            achieved_mw += data['contracted_dr_mw']
            data['active'] = True
    return achieved_mw
```

## 3.5 GRIDCo Fault Watch Agent — Purely Reactive

**Agent Profile — gridco_fault_watch@gridmind.gh**

```
Type:        Purely Reactive Agent — no deliberation whatsoever
JID:         gridco_fault_watch@gridmind.gh
Institution: Models GRIDCo's System Control Centre monitoring function
Behaviour:   MessageReceiveBehaviour (stateless, zero-latency response)
```

GRIDCo Fault Watch is the system's clearest example of a purely reactive agent and a deliberate architectural contrast to ECG Central Dispatch. It has no goal, no plan library, and no internal deliberation. Its sole function is: receive district INFORM, check statistical threshold, fire alert if threshold breached. This agent exists to demonstrate to the evaluator the architectural distinction between reactive and deliberative agents — a core Prometheus concept.

### Anomaly Detection Rules

```
# Rule 1: Demand Spike (upward anomaly)
if current_mw > mean(last_5_readings) + FAULT_SIGMA * stdev(last_5_readings):
    fault_type = 'demand_spike'
    severity   = (current_mw - mean(last_5)) / stdev(last_5)
    send INFORM(to='ecg_central_dispatch@gridmind.gh',
                content={'district': district_id, 'fault_type': fault_type,
                         'severity_sigma': severity, 'tick': tick})

# Rule 2: Feeder Loss (sudden zero-reading anomaly)
if current_mw < mean(last_5_readings) * 0.15 and mean(last_5_readings) > 50:
    fault_type = 'feeder_loss'  # e.g. Spintex Road corridor fault
    send INFORM(to='ecg_central_dispatch@gridmind.gh',
                content={'district': district_id, 'fault_type': 'feeder_loss',
'tick': tick})
```

## 3.6 ECG Forecast Unit Agent

**Agent Profile — ecg_forecast_unit@gridmind.gh**

```
Type:        Deliberative Advisory Agent
JID:         ecg_forecast_unit@gridmind.gh
Institution: Models ECG's Load Forecasting Unit (a real department within ECG)
Behaviour:   PeriodicBehaviour (fires every FORECAST_INTERVAL ticks)
```

### Forecasting Model

The ECG Forecast Unit uses a weighted moving average over each district's last 20-tick demand history. Recent readings are weighted more heavily to capture trend momentum. The model is intentionally simple — the point is the agent architecture, not ML sophistication:

```
WEIGHTS = [0.03, 0.04, 0.04, 0.05, 0.05, 0.06, 0.06, 0.07, 0.07, 0.08,
           0.08, 0.07, 0.07, 0.06, 0.06, 0.05, 0.05, 0.04, 0.04, 0.03]
```

```
# sum(WEIGHTS) = 1.0, recent ticks (index 9-12) weighted highest

def forecast_district(self, district_id: str) -> float:
    history = self.beliefs['demand_history'][district_id][-20:]
    if len(history) < 5:
        return history[-1] if history else 0.0
    weights = WEIGHTS[-len(history):]
    weights = [w / sum(weights) for w in weights]  # renormalise
    return sum(w * r for w, r in zip(weights, history))
```

## Advisory Logic

```
for district_id, forecasted_mw in self.forecast_all_districts().items():
    ceiling_mw = DISTRICT_CAPACITY[district_id]
    if forecasted_mw > ceiling_mw * PREEMPTIVE_THRESHOLD:
        send INFORM(to='ecg_central_dispatch@gridmind.gh', content={
            'district':      district_id,
            'forecast_mw':   forecasted_mw,
            'ceiling_mw':    ceiling_mw,
            'utilisation':   forecasted_mw / ceiling_mw,
            'horizon_ticks': FORECAST_HORIZON,
            'confidence':    self.compute_confidence(district_id)
        })
```

# 4. FIPA-ACL Communication Design

## 4.1 Performative Reference

| Performative | Direction | Ghana Grid Meaning |
|---|---|---|
| INFORM | Zone agents → ecg_central_dispatch, gridco_fault_watch | District demand update — the backbone of the system's situational awareness |
| INFORM | gridco_fault_watch → ecg_central_dispatch | Anomaly alert: demand spike (Tema shift-change) or feeder loss (Spintex Road) |
| INFORM | ecg_forecast_unit → ecg_central_dispatch | Predicted demand surge — triggers preemptive ECG response before the spike arrives |
| INFORM | renewable agent → ecg_central_dispatch | Injection confirmation: 'Kaleo Solar now injecting 120 MW into Volta-Accra 161kV line' |
| INFORM | accra_industrial_dr → ecg_central_dispatch | DR result: 'VALCO and Tema Oil Refinery have reduced by 110 MW total' |
| CFP | ecg_central_dispatch → kaleo_solar / nzema_solar / keta_wind | Call for energy proposals: ECG Dispatch requests bids from available renewable sources |
| PROPOSE | renewable agent → ecg_central_dispatch | Energy bid: available MW and cost coefficient in response to ECG CFP |
| REFUSE | renewable agent → ecg_central_dispatch | No capacity: 'Kaleo Solar reporting near-zero output — cloud cover event' |
| ACCEPT-PROPOSAL | ecg_central_dispatch → winning renewable | Procurement award: ECG Dispatch selects the best renewable bid |
| REJECT-PROPOSAL | ecg_central_dispatch → losing renewables | Bid rejected: other sources were more cost-effective or higher capacity |
| REQUEST | ecg_central_dispatch → accra_industrial_dr | Demand reduction order: ECG activates industrial DR programme |
| AGREE | accra_industrial_dr → ecg_central_dispatch | DR acknowledgement: Accra Industrial DR confirms it has received the REQUEST |
| FAILURE | accra_industrial_dr → ecg_central_dispatch | DR shortfall: target MW reduction could not be fully achieved from the roster |
| BROADCAST | ecg_central_dispatch → all zone agents | System digest: current grid status, active intention, and audit reference |

## 4.2 Contract Net Protocol — Ghana Grid Energy Procurement Trace

The following is the complete message trace for a successful Contract Net procurement round triggered by Scenario 2: Tema Shift-Change Spike. This is the most important communication pattern in GridMind:

**Complete CNP Trace — Tema 18:00 Shift-Change Spike**

```
Tick 47:  tema_industrial sends INFORM: { demand_mw: 398, capacity_mw: 400,
trend: 'rising' }
          gridco_fault_watch detects 3.2σ spike → sends INFORM to
ecg_central_dispatch:
          { district: 'tema_industrial', fault_type: 'demand_spike',
severity_sigma: 3.2 }

Tick 48:  ecg_central_dispatch reconsiders intention: MONITOR →
RESPOND_TO_FAULT
          Generates conversation_id: 'cnp-gh-20240315-048'
          Sends CFP to kaleo_solar, nzema_solar, keta_wind:
          { request_mw: 85.0, deadline_ticks: 3, conversation_id: 'cnp-gh-
20240315-048' }

Tick 49a: kaleo_solar responds PROPOSE:
          { source: 'kaleo_solar', available_mw: 110.0, cost_coeff: 0.28,
            conversation_id: 'cnp-gh-20240315-048' }
          (it is 17:45 local time — still high solar output)

Tick 49b: nzema_solar responds PROPOSE:
          { source: 'nzema_solar', available_mw: 18.0, cost_coeff: 0.31,
            conversation_id: 'cnp-gh-20240315-048' }

Tick 49c: keta_wind responds PROPOSE:
          { source: 'keta_wind', available_mw: 42.0, cost_coeff: 0.19,
            conversation_id: 'cnp-gh-20240315-048' }

Tick 50:  ecg_central_dispatch evaluates proposals (score = available_mw /
cost_coeff):
          kaleo_solar: 110.0 / 0.28 = 392.9  ← WINNER
          keta_wind:    42.0 / 0.19 = 221.1
          nzema_solar:  18.0 / 0.31 =  58.1
          Sends ACCEPT-PROPOSAL to kaleo_solar@gridmind.gh
          Sends REJECT-PROPOSAL to nzema_solar and keta_wind

Tick 51:  kaleo_solar sends INFORM confirm:
          { injecting_mw: 85.0, source: 'kaleo_solar', tick: 51 }
          ecg_central_dispatch updates beliefs, logs CNP trace to audit_log
          Desire falls back to MONITOR — Tema Industrial returns to 93%
utilisation
```

## 4.3 FIPA-ACL Message Envelope Schema

All messages use a consistent envelope. The ontology field locks messages to GridMind's schema version, preventing cross-system contamination in multi-tenant XMPP environments:

```
from spade.message import Message
from uuid import uuid4
import json

def build_message(to_jid: str, performative: str, payload: dict,
                  conversation_id: str = None) -> Message:
    msg = Message(to=to_jid)
    msg.set_metadata('performative',    performative)
    msg.set_metadata('ontology',        'gridmind-ecg-gridco-v1')
    msg.set_metadata('language',        'fipa-sl')
    msg.set_metadata('conversation-id', conversation_id or str(uuid4()))
    msg.body = json.dumps({
        'sender':    str(self.jid),
        'tick':      self.env.tick,
        **payload
    })
    return msg
```

# 5. System Architecture & Project Structure

## 5.1 Directory Layout

```
gridmind/
├── agents/
│   ├── __init__.py
│   ├── zone_agent.py              # Base class — all 5 district agents inherit from
this
│   ├── ecg_central_dispatch.py    # BDI showpiece agent
│   ├── renewable_source_agent.py  # Base class — kaleo, nzema, keta inherit
│   ├── kaleo_solar_agent.py
│   ├── nzema_solar_agent.py
│   ├── keta_wind_agent.py
│   ├── demand_response_agent.py   # accra_industrial_dr
│   ├── fault_watch_agent.py       # gridco_fault_watch
│   └── forecast_unit_agent.py     # ecg_forecast_unit
├── bdi/
│   ├── belief_base.py             # ECGCentralDispatchBeliefs dataclass + update()
│   ├── desire_evaluator.py        # Priority-ordered desire condition evaluation
│   └── intention_selector.py      # Intention → plan resolution + reconsideration
├── environment/
│   ├── ghana_grid_state.py        # GhanaGridEnvironment dataclass
│   ├── event_injector.py          # Named scenario event scheduler
│   └── scenarios/
│       ├── akosombo_curtailment.py
│       ├── tema_shift_spike.py
│       ├── spintex_feeder_fault.py
│       ├── gridco_line_trip.py
│       └── dumsor_rotation.py
├── communication/
│   ├── message_factory.py         # build_message() with ECG/GRIDCo envelope
│   └── contract_net.py            # CNP orchestration: CFP → collect → evaluate →
award
├── dashboard/
│   └── ecg_terminal_dashboard.py  # Rich-library live dashboard
├── evaluation/
│   ├── metrics_collector.py
│   └── ecg_baseline_runner.py     # Threshold-only comparator
├── tests/
│   ├── test_bdi_dispatch.py
│   ├── test_contract_net.py
│   ├── test_fault_detection.py
│   └── test_forecast_unit.py
├── main.py                        # Boot sequence + simulation driver
├── config.py                      # All Ghana-specific parameters
└── requirements.txt
```

## 5.2 Dependencies

| Package | Version | Purpose |
|---|---|---|
| spade | >=3.2.0 | Core XMPP-based multi-agent framework — all agent lifecycle and messaging |
| asyncio | stdlib | Async behaviour execution within SPADE's event loop |
| numpy | >=1.24 | Statistical computations in GRIDCo Fault Watch and ECG Forecast Unit |
| rich | >=13.0 | Terminal dashboard — live ECG grid status display |
| aioxmpp | auto with SPADE | Underlying XMPP transport layer |
| pytest + pytest-asyncio | latest | Unit and integration testing of agent logic |
| matplotlib | optional | Post-run KPI visualisation (baseline vs. MAS comparison charts) |

## 5.3 Agent Boot Sequence

Agents are started in dependency order to prevent message loss during initialisation. Listeners must be online before senders begin broadcasting:

1. Start Prosody XMPP server (Docker, already configured in Codespaces devcontainer.json)
2. Start gridco_fault_watch@gridmind.gh — must be ready before district agents send INFORM
3. Start ecg_forecast_unit@gridmind.gh — listener ready before zone data flows
4. Start all five district ZoneAgents — tema_industrial, accra_central, kumasi_suame, kasoa_corridor, takoradi_harbour
5. Start kaleo_solar, nzema_solar, keta_wind — renewable agents ready before CNP is possible
6. Start accra_industrial_dr — DR agent ready before emergency escalation
7. Start ecg_central_dispatch@gridmind.gh last — all its message targets are now live
8. Launch ecg_terminal_dashboard and event_injector coroutine
9. Run simulation for configured tick count or until KeyboardInterrupt

# 6. BDI Implementation — ECG Central Dispatch

This section provides implementation-level code sketches for the BDI architecture of ecg_central_dispatch@gridmind.gh. This agent is the centerpiece of the GridMind system and should be the most thoroughly demonstrated during the live evaluation.

## 6.1 Belief Update with Audit Trail

```
# belief_base.py
from dataclasses import dataclass, field
from typing import Dict, List, Optional

@dataclass
class ECGCentralDispatchBeliefs:
    district_demands: Dict[str, float] = field(default_factory=lambda: {
        'tema_industrial': 0.0,  'accra_central': 0.0,
        'kumasi_suame':    0.0,  'kasoa_corridor': 0.0,
        'takoradi_harbour': 0.0
    })
    total_managed_capacity_mw: float = 1015.0
    renewable_bids: Dict[str, float] = field(default_factory=lambda: {
        'kaleo_solar': 0.0, 'nzema_solar': 0.0, 'keta_wind': 0.0
    })
    active_faults: List[str]    = field(default_factory=list)
    forecast: Dict[str, float]  = field(default_factory=dict)
    audit_log: List[dict]       = field(default_factory=list)

    def update(self, key: str, value, reason: str, tick: int):
        old_val = getattr(self, key, None)
        setattr(self, key, value)
        self.audit_log.append({
            'tick': tick, 'agent': 'ecg_central_dispatch',
            'belief_updated': key, 'old': old_val, 'new': value,
            'reason': reason
        })
```

## 6.2 Desire Evaluator

```
# desire_evaluator.py
DESIRES = [
    ('EMERGENCY_STABILISE',
     lambda b: sum(b.district_demands.values()) > b.total_managed_capacity_mw *
0.98),

    ('RESPOND_TO_FAULT',
     lambda b: len(b.active_faults) > 0),
```

```
    ('PREEMPTIVE_RESPONSE',
     lambda b: any(
         fcast_mw > DISTRICT_CAPACITY[d] * 0.85
         for d, fcast_mw in b.forecast.items()
     )),

    ('OPTIMISE_RENEWABLES',
     lambda b: sum(b.renewable_bids.values()) > 0
               and is_demand_rising(b.district_demands)),

    ('MONITOR', lambda b: True),  # always-true fallback
]

def evaluate_desires(beliefs) -> str:
    for desire_name, condition in DESIRES:
        if condition(beliefs):
            return desire_name
```

## 6.3 BDI Cycle — Intention Reconsideration Loop

```
# In ecg_central_dispatch.py
class BDICycleBehaviour(CyclicBehaviour):
    async def run(self):
        await self.process_all_incoming_messages()   # updates beliefs

        new_desire = evaluate_desires(self.agent.beliefs)

        if new_desire != self.agent.current_desire:
            # ── INTENTION RECONSIDERATION ──────────────────────────
            old = self.agent.current_desire
            self.agent.beliefs.audit_log.append({
                'tick': self.agent.env.tick,
                'event': 'INTENTION_RECONSIDERATION',
                'from_desire': old, 'to_desire': new_desire,
            })
            await self.abandon_current_intention(old)
            self.agent.current_desire = new_desire
            await self.adopt_new_intention(new_desire)
        else:
            await self.continue_current_plan()

        await asyncio.sleep(TICK_INTERVAL)
```

# 7. Evaluation Strategy

GridMind's evaluation compares two modes running identical event sequences: GridMind MAS mode (all 12 agents active, full BDI + CNP + DR) versus ECG Baseline mode (threshold-only rule: if demand > 90% capacity, immediately shed lowest-priority district). The comparison produces hard numbers across five KPIs.

## 7.1 Key Performance Indicators

| KPI | Definition | Measurement Method |
|---|---|---|
| Dumsor Frequency | Number of ticks where any district enters 'shedding' status per 200-tick run | Count env.district_status == 'shedding' events |
| Shedding Duration | Total tick-duration of load shedding across all districts per run | Sum of consecutive 'shedding' ticks per district |
| Response Latency | Ticks between GRIDCo Fault Watch INFORM and first corrective action | Timestamp delta in audit_log |
| Renewable Utilisation | Actual MW procured from Kaleo/Nzema/Keta as % of total available across the run | Sum(injected_mw) / Sum(available_mw) per source |
| DR Success Rate | % of DR requests where accra_industrial_dr achieved ≥ 90% of target MW | INFORM achieved_mw / REQUEST target_mw across all DR activations |
| Forecast Accuracy | Mean Absolute Error (MAE) of ecg_forecast_unit predictions vs. actual demand | Mean(\|forecast_mw - actual_mw\|) per district per tick |
| Audit Completeness | % of decisions in audit_log that have a complete reason chain (belief → desire → intention → plan) | Structural check on audit_log entries |

## 7.2 ECG Baseline Mode

The baseline runner replaces ecg_central_dispatch with a simple threshold function. It receives the same district INFORM messages and applies one rule only: when total managed demand exceeds 90% of 1,015 MW (i.e., > 913.5 MW), it immediately sheds kasoa_corridor (residential first). When demand exceeds 95%, it also sheds accra_central. No renewable procurement, no demand response, no forecasting. This represents ECG's current operational approach to load management.

## 7.3 Test Scenario Evaluation Matrix

| Scenario | MAS Advantage Expected | Baseline Behaviour | Key Metric |
|---|---|---|---|
| Akosombo Low-Water Curtailment | Forecast Unit detects supply erosion; preemptive Keta Wind procurement activated before peak hours | Waits for threshold breach; sheds Kasoa immediately | Shedding Duration |
| Tema 06:00 / 18:00 Shift-Change Spike | GRIDCo Fault Watch fires within 1 tick; Kaleo Solar procured via CNP; no shedding | Sheds Kasoa within 2 ticks of spike; no renewable option explored | Dumsor Frequency + Renewable Utilisation |
| Dumsor Rotation Forced Activation | Prioritised shedding (Kasoa first) with full audit trace and reason chain | Unstructured immediate shedding; no audit | Audit Completeness |
| Spintex Road Feeder Fault | Dispatch agent isolates faulty zone belief; continues managing remaining 4 districts correctly | Faulty reading triggers incorrect shedding cascade | Response Latency |
| GRIDCo Kumasi Line Trip | Cascading belief revision: Kumasi capacity ceiling revised; DR activated for Western Region consumers | Manual intervention required; baseline has no capacity belief model | DR Success Rate |

## 7.4 Projected KPI Comparison

### 📊 Projected Results — GridMind MAS vs. ECG Baseline (200-tick run, 5 scenarios)

Dumsor Frequency: MAS ~65% fewer shedding events than baseline
Shedding Duration: MAS ~55% shorter total shedding time per run
Renewable Utilisation: MAS ~4× higher (active CNP vs. zero renewable engagement in baseline)
Response Latency: MAS: 1–2 ticks (reactive faults) | 0 ticks (predicted surges via Forecast Unit)
Baseline: 2–4 ticks (polling delay only)
DR Success Rate: Expected ~87% across 20-run evaluation sweep
Forecast MAE: Expected ±8–15 MW per district (weighted MA is conservative)
Audit Completeness: MAS: 100% (by design) | Baseline: 0% (no audit mechanism)

# 8. Live Demonstration Script

This script is written for a 10–12 minute live evaluation demo. Each step targets a specific course concept. The evaluator should be walked through each step with the exact talking points provided below.

### 🖥️ Terminal Layout (3-pane setup)

Left pane: FIPA-ACL message log — every performative, scrolling in real-time
Centre pane: ECG Grid Dashboard (Rich library) — 5 district status tiles, intention display, KPI counters
Right pane: Event injector REPL — manual scenario injection

Prerequisite: All 12 agents running via python main.py --demo-mode
demo-mode slows TICK_INTERVAL to 1.5s for observability

## Step 1 — Normal Operations (1–2 min)

Talking point: "GridMind is live. You are looking at a multi-agent proof-of-concept for ECG and GRIDCo — a system designed to autonomously prevent dumsor. Right now, all five districts are within safe operating range. ECG Central Dispatch is in its MONITOR intention — its lowest-priority desire. Watch the INFORM messages flowing left: those are tema_industrial, accra_central, kumasi_suame, kasoa_corridor, and takoradi_harbour reporting demand every tick."

- Point to message log: INFORM performatives from 5 districts to ecg_central_dispatch
- Point to dashboard: all 5 district tiles GREEN, active_intention = MONITOR
- Point to belief base panel: district_demands updating each tick with realistic MW values
  - e.g. tema_industrial: 293 MW / 400 MW (73%) — Accra Central: 142 MW / 190 MW (75%)

## Step 2 — ECG Forecast Unit Fires (2 min)

Talking point: "The ECG Forecast Unit — modelling ECG's real Load Forecasting Department — has been building 20-tick rolling demand histories for all five districts. Watch what happens when it projects the Tema shift-change approaching."

- Inject: inject_event('tema_gradual_ramp', rate=1.04, duration=8)
- Watch ecg_forecast_unit compute rising forecast for tema_industrial
- Forecast crosses 85% of 400 MW ceiling → ecg_central_dispatch receives INFORM from forecast unit
- Desire changes: MONITOR → PREEMPTIVE_RESPONSE (visible in dashboard intention panel)
- CNP fires before the actual spike materialises: CFP sent to all 3 renewable agents
- Kaleo Solar wins the bid (110 MW available at 17:45 local time) → ACCEPT-PROPOSAL visible in message log

Talking point: "No shedding. The system acted on a prediction, not a crisis. That is the value of deliberative BDI over a reactive threshold rule."

## Step 3 — Intention Reconsideration (2 min)

Talking point: "This is the BDI moment. I am going to inject a sudden feeder fault on the Spintex Road corridor — ECG's Accra Central feeder — while ECG Central Dispatch is mid-plan. Watch what happens."

- While the renewable optimisation plan is running: inject_event('spintex_feeder_fault')
- gridco_fault_watch detects sudden near-zero reading from accra_central → sends INFORM { fault_type: 'feeder_loss' }
- In the audit log panel: INTENTION_RECONSIDERATION fires — from_desire: OPTIMISE_RENEWABLES, to_desire: RESPOND_TO_FAULT
- ecg_central_dispatch isolates accra_central from its belief base and continues managing the other 4 districts

Talking point: "The agent stopped what it was doing, re-evaluated its priorities, and adopted a new intention. That is intention reconsideration — the cognitive mechanism that distinguishes a BDI agent from a simple rule system."

## Step 4 — Contract Net Under Pressure (2 min)

Talking point: "Now let me stress the system. I will inject the Akosombo low-water curtailment scenario — VRA reduces Akosombo output — while simultaneously triggering the 18:00 Tema shift-change spike."

- inject_event('akosombo_curtailment', reduction_pct=0.40)
- inject_event('tema_shift_spike', hour=18, multiplier=1.22)
- Tema demand hits 389 MW — gridco_fault_watch fires a 3.1σ anomaly alert
- ecg_central_dispatch escalates to EMERGENCY_STABILISE desire
- CNP fires: CFP to kaleo_solar, nzema_solar, keta_wind
- It is now 18:00 — Kaleo Solar output drops to ~55 MW (evening). Keta Wind at 38 MW
- ecg_central_dispatch accepts Kaleo + Keta combined. Still 45 MW short.
- Falls through to demand_response_plan: REQUEST sent to accra_industrial_dr
- VALCO (80 MW contracted) and Tema Oil Refinery (35 MW) engaged → 110 MW offered → 45 MW gap closed
- No shedding. Audit log records the full decision chain.

## Step 5 — Forced Dumsor Rotation (1–2 min)

Talking point: "Let me push past all available options. I will remove all renewable capacity and exhaust the DR roster — forcing a controlled dumsor rotation."

- inject_event('all_renewables_offline')  # weather + night-time combined
- inject_event('demand_surge_all_zones', multiplier=1.18)
- CNP returns all REFUSE messages — log visible

- DR agent returns FAILURE with gap_mw = 80 — insufficient roster remaining
- ecg_central_dispatch activates load_shedding_plan
- Shedding order: kasoa_corridor first (residential), then accra_central if needed
- Tema Industrial and Takoradi Harbour protected — port and industry last resort
- Dashboard shows Kasoa tile turning RED with reason_code: DUMSOR_ROTATION

Talking point: "Even the worst case — a forced dumsor rotation — is handled with a prioritised, auditable decision. Every choice has a reason code. This is the difference between intelligent load management and arbitrary shedding."

## Step 6 — Baseline Comparison (1 min)

Talking point: "Finally, let me run the same scenario sequence through the ECG baseline — a threshold-only rule with no renewable procurement, no demand response, no forecasting."

- python evaluation/ecg_baseline_runner.py --scenario full_stress_sequence
- Show side-by-side terminal output: baseline sheds Kasoa at Tick 3; MAS never sheds in the same scenario
- Show KPI table: dumsor_frequency, shedding_duration_ticks, renewable_utilisation_pct

Talking point: "The numbers are the argument. GridMind is what intelligent agent coordination looks like when it is applied to a problem that Ghana has been living with for twenty years."

# 9. Implementation Phases

Phases are ordered by technical dependency — complete each phase before starting the next. Time allocation is a guide only; prioritise correctness over schedule.

## Phase 1 — Infrastructure & Environment

- Configure GitHub Codespaces devcontainer.json with Prosody XMPP Docker service
- Install SPADE, verify two test agents exchange a hello-world INFORM successfully
- Implement ghana_grid_state.py: GhanaGridEnvironment dataclass with all 5 district fields
- Implement config.py with all Ghana-specific parameters (see Section 10)
- Implement message_factory.py: build_message() with ECG/GRIDCo ontology envelope

## Phase 2 — Reactive Foundation Agents

- Implement ZoneAgent base class with PeriodicBehaviour, belief structure, and INFORM sending
- Instantiate all 5 district agents with per-district capacity and noise configs
- Implement gridco_fault_watch with rolling-window spike and feeder_loss detection
- Integration test: tema_industrial spike triggers gridco_fault_watch INFORM within 1 tick

## Phase 3 — ECG Central Dispatch BDI Core

- Implement ECGCentralDispatchBeliefs dataclass with update() audit trail
- Implement desire_evaluator.py with all 5 DESIRES in priority order
- Implement BDILoopBehaviour with intention reconsideration
- Implement monitoring_plan and passive_monitoring
- Unit test: belief updates correctly, EMERGENCY_STABILISE desire activates at 98% threshold

## Phase 4 — Renewable Agents & Contract Net

- Implement RenewableSourceAgent base class with capacity simulation and CFP response
- Instantiate kaleo_solar (155 MW ceiling), nzema_solar (20 MW), keta_wind (50 MW)
- Implement contract_net.py: CFP broadcast → proposal collection → scoring → award
- Integration test: full CNP round completes within CNP_DEADLINE_TICKS

## Phase 5 — Demand Response & Forecasting

- Implement accra_industrial_dr with VALCO, Tema Oil Refinery, Meridian Port, GHACEM, Accra Brewery roster
- Implement compliance-probability negotiation logic
- Implement ecg_forecast_unit with 20-tick weighted moving average per district
- Wire ForecastAgent output into PREEMPTIVE_RESPONSE desire evaluation
- End-to-end test: Forecast Unit predicts Tema spike → CNP fires preemptively → no shedding

## Phase 6 — Named Scenarios & Dashboard

- Implement all 5 named scenario event injectors in environment/scenarios/
- Implement ecg_terminal_dashboard.py with Rich: district tiles, intention panel, FIPA log, KPI counters
- Implement ecg_baseline_runner.py: threshold-only ECG comparator
- Run all 5 scenarios, record KPI table for evaluation

## Phase 7 — Prometheus Documentation & Tests

- Finalise this implementation plan document with actual measured KPI values
- Draw the Prometheus architectural design interaction diagram (tools: draw.io or hand-drawn)
- Write per-agent detailed design tables for Prometheus submission
- Write unit tests: test_bdi_dispatch.py, test_contract_net.py, test_fault_detection.py, test_forecast_unit.py

# 10. Configuration Reference — config.py

No placeholder values. Every parameter here is grounded in Ghana's real grid infrastructure or derived from it:

```
# ─── GridMind Configuration ──────────────────────────────────────────────
# Proof-of-concept pilot: ECG + GRIDCo Distributed Load Management System
# University of Ghana — DCIT 403

# ─── XMPP Server ─────────────────────────────────────────────────────────
XMPP_SERVER   = 'gridmind.gh'          # Prosody domain (configured in Codespaces)
XMPP_PORT     = 5222

# ─── Agent JIDs ──────────────────────────────────────────────────────────
ECG_DISPATCH_JID    = 'ecg_central_dispatch@gridmind.gh'
GRIDCO_FAULT_JID    = 'gridco_fault_watch@gridmind.gh'
ECG_FORECAST_JID    = 'ecg_forecast_unit@gridmind.gh'
ACCRA_DR_JID        = 'accra_industrial_dr@gridmind.gh'
KALEO_SOLAR_JID     = 'kaleo_solar@gridmind.gh'
NZEMA_SOLAR_JID     = 'nzema_solar@gridmind.gh'
KETA_WIND_JID       = 'keta_wind@gridmind.gh'

DISTRICT_JIDS = [
    'tema_industrial@gridmind.gh',
    'accra_central@gridmind.gh',
    'kumasi_suame@gridmind.gh',
    'kasoa_corridor@gridmind.gh',
    'takoradi_harbour@gridmind.gh',
]

# ─── District Capacity Ceilings (MW) ─────────────────────────────────────
# Based on ECG load data and district economic characterisation
DISTRICT_CAPACITY = {
    'tema_industrial':  400.0,  # Heavy industry, Meridian Port, TOR, VALCO
    'accra_central':    190.0,  # Government, CBD, hospitals, Ridge
    'kumasi_suame':     150.0,  # Auto/metal industry, Suame Magazine
    'kasoa_corridor':   100.0,  # High-density residential, small commerce
    'takoradi_harbour': 175.0,  # Port ops, oil support, GHACEM
}
TOTAL_MANAGED_CAPACITY_MW = 1015.0  # Sum of pilot districts

# ─── District Base Demand (MW) — Normal Operations ───────────────────────
DISTRICT_BASE_DEMAND = {
    'tema_industrial':  290.0,
    'accra_central':    140.0,
    'kumasi_suame':     110.0,
    'kasoa_corridor':    65.0,
    'takoradi_harbour': 125.0,
}
```

```
# ─── Renewable Source Capacities (MW) ──────────────────────────────────────
# Kaleo: 155MW installed (Wa, Upper West) — BXC / GoG
# Nzema: ~20MW operational (Atuabo, Western Region) — Blue Energy
# Keta Wind: ~50MW modelled (Accra-Keta coastal corridor)
RENEWABLE_MAX_CAPACITY = {
    'kaleo_solar':  155.0,
    'nzema_solar':   20.0,
    'keta_wind':     50.0,
}
RENEWABLE_COST_COEFFICIENTS = {
    'kaleo_solar': 0.28,  # Low marginal cost (utility solar)
    'nzema_solar': 0.31,  # Slightly higher (smaller plant, older PPA)
    'keta_wind':   0.19,  # Lowest marginal cost (wind near zero marginal)
}

# ─── Demand Response Roster (MW contracted) ────────────────────────────────
DR_CONSUMER_ROSTER = {
    'VALCO':              {'contracted_dr_mw': 80,  'compliance_rate': 0.95},
    'tema_oil_refinery':  {'contracted_dr_mw': 35,  'compliance_rate': 0.88},
    'meridian_port':      {'contracted_dr_mw': 20,  'compliance_rate': 0.80},
    'GHACEM_takoradi':    {'contracted_dr_mw': 25,  'compliance_rate': 0.82},
    'accra_brewery':      {'contracted_dr_mw': 15,  'compliance_rate': 0.75},
}
TOTAL_DR_CONTRACTED_MW = 175.0

# ─── BDI Thresholds ────────────────────────────────────────────────────────
EMERGENCY_THRESHOLD_PCT   = 0.98  # EMERGENCY_STABILISE desire activates
PREEMPTIVE_THRESHOLD_PCT  = 0.85  # PREEMPTIVE_RESPONSE desire activates
OPTIMISE_THRESHOLD_PCT    = 0.70  # OPTIMISE_RENEWABLES desire activates

# ─── Simulation Parameters ─────────────────────────────────────────────────
TICK_INTERVAL         = 0.5    # seconds (set 1.5 for demo mode)
SIMULATION_TICKS      = 200
CNP_DEADLINE_TICKS    = 3      # ticks to wait for all PROPOSE messages
FORECAST_HORIZON      = 5      # ticks ahead to project
FORECAST_HISTORY_LEN  = 20     # rolling history per district
FAULT_WINDOW          = 5      # rolling window for anomaly detection
FAULT_SIGMA_THRESHOLD = 2.0    # standard deviations for spike detection
BROADCAST_INTERVAL    = 5      # ticks between system digest broadcasts
```

# 11. Edge Cases & Risk Mitigations

| Edge Case / Risk | Real-World Ghana Analogue | Mitigation |
|---|---|---|
| All 3 renewable agents send REFUSE in same CNP round | Cloudy dry-season day: Kaleo and Nzema at ~5% output; calm coastal air kills Keta Wind | Fall through immediately to demand_response_plan after collecting all REFUSEs or after CNP_DEADLINE_TICKS expire |
| DR agent achieves less than 50% of target MW | VALCO is already at minimum operations; Tema Oil Refinery maintenance outage | Return FAILURE with gap_mw field; ecg_central_dispatch activates partial load_shedding_plan for the gap only |
| Spintex Road feeder fault misread as genuine demand drop | ECG feeder cable faults on Spintex Road are historically common | gridco_fault_watch uses Rule 2 (feeder_loss) when reading drops to < 15% of rolling mean; dispatches different fault_type to ecg_central_dispatch for belief isolation rather than shedding |
| ecg_central_dispatch receives overlapping CNP conversation IDs | Two simultaneous rapid spikes in different districts | All proposals are keyed by conversation_id in pending_cnp_proposals; concurrent CNPs are queued — second CNP fires after first ACCEPT-PROPOSAL is sent |
| ecg_forecast_unit sends false-positive surge warning | Temporary equipment noise on Tema Industrial sensor | PREEMPTIVE_RESPONSE triggers a soft CNP (smaller request_mw); cost of false positive is minor over-procurement, not shedding. Logged as 'preemptive_false_positive' in audit_log |
| XMPP message delivery delay under high agent load | Prosody server congestion during multi-scenario stress test | All outgoing messages have a 2-tick retry on timeout; SPADE behaviours use async/await throughout — no blocking I/O |
| GRIDCo line trip isolates Kumasi from Accra transmission | GRIDCo Volta-Kumasi 161kV trips are historically documented | kumasi_suame agent updates its capacity_ceiling_mw belief downward; ecg_central_dispatch receives the revised INFORM and cascades to DR request for Ashanti Region consumers |
| Simulation tick interval too fast for demo observability | N/A — demo environment issue | TICK_INTERVAL = 1.5 in demo-mode flag; all 12 agents support graceful slow-down without message ordering issues |